

QimenBench: Cross-Source Target-Control Evidence for Procedural Chart Construction

Anonymous submission

Abstract

Procedural chart-construction tasks can be easy to score yet hard to evaluate honestly: a system may copy visible fields while failing the hidden derivation that the task was meant to test. We introduce QimenBench, a paired target-control benchmark for Qi Men Dun Jia chart fields, and add an independent procedural-source validation packet covering Gregorian calendar arithmetic, Luhn checksum arithmetic, and Roman numeral transduction. Each source uses visible-answer controls and hidden-derivation targets with identical output schemas, allowing exact target-control comparison rather than a single aggregate score. In QimenBench v0.1.0, the oracle sanity condition matches 1,440/1,440 scored rows; extraction and hosted GPT-5 mini each match 720/1,440 rows by solving controls and failing hidden targets; and Qwen2.5-7B-Instruct matches 18/1,440 rows. In the independent source packet, oracle rows match 72/72 tasks, while a visible-answer extractor matches 36/72 by solving all controls and no hidden targets. The cross-source matrix supports a bounded benchmark-construction claim: target-control pairing exposes visible-copying shortcuts across independently sourced procedural rules.

Introduction

Benchmark construction is increasingly constrained by saturation, leakage, and shortcut behavior rather than by a shortage of leaderboards. Benchmark-saturation studies argue that evaluation sets lose value when they reward memorized patterns or superficial task cues instead of the intended capability (Akhtar et al. 2026; Deveci and Ataman 2025). Multi-modal benchmark papers such as MMMU and SEED-Bench respond with explicit task families, provenance, and scoring contracts that make the evaluated slice inspectable (Yue et al. 2023; Li et al. 2023). Agent and environment-facing benchmarks add longer-horizon interaction, but their breadth can make it hard to tell whether a failure is caused by planning, tool use, parsing, or a missing domain procedure (Liu et al. 2023a; Wang et al. 2023a). General reasoning evaluations and HELM-style reporting broaden measurement coverage, yet aggregate final-answer scores still often hide whether a system used the intended derivation path (Srivastava et al. 2022; Liang et al. 2022).

QimenBench studies a narrower diagnostic problem: exact procedural chart construction with paired visible controls and hidden-derivation targets. The benchmark uses Qi

Men Dun Jia chart fields because they create compact structured outputs with visible components and hidden procedural rules. A model or heuristic can copy a visible control answer, but target rows require deriving fields that are not directly present in the prompt. The target-control pair is therefore the central experimental unit rather than a convenience split.

The completed evidence packet is deliberately bounded and cross-source. It contains full 1,440-row QimenBench oracle, extraction-only, hosted GPT-5 mini, and Qwen2.5-7B-Instruct runs, plus a 72-row independent procedural-source packet with oracle and visible-answer extractor baselines. The QimenBench oracle reaches 1,440/1,440 exact rows, establishing scorer/gold consistency. Extraction and hosted GPT-5 mini each reach 720/1,440 exact rows by solving 720/720 controls and 0/720 hidden targets. The Qwen condition reaches 18/1,440 rows. The independent-source oracle reaches 72/72 exact rows, while the shortcut extractor reaches 36/72 by solving all controls and no hidden targets.

The source boundary is explicit. QimenBench F1/F2/F3 have distinct task files, constructs, schemas, raw predictions, scored rows, counts, license/access notes, and rationale, but they share the same project-created rule manifest and provenance. The added procedural packet supplies three independent rule sources with separate provenance: public-domain Gregorian calendar arithmetic, public Luhn checksum arithmetic, and public-domain Roman numeral notation. These sources validate the target-control measurement pattern; they do not turn the paper into a broad model leaderboard.

The design makes negative evidence useful. In a conventional benchmark report, 0.500 accuracy might be dismissed as a weak baseline result. In QimenBench, the same aggregate decomposes into a precise behavioral signature: controls are exact and hidden targets fail. That signature identifies visible-answer copying rather than noisy half-success, so the paper treats the target-control gap, not the aggregate score, as the main measurement object.

This paper contributes three bounded artifacts. First, it specifies a benchmark construction pattern for paired procedural tasks. Second, it reports a frozen QimenBench v0.1.0 packet with oracle, extraction, hosted, and open-weight rows. Third, it reports an independent-source validation matrix showing that the same shortcut signature appears in Gre-

gorian calendar, Luhn checksum, and Roman numeral procedures under oracle and visible-extraction baselines. The model rows remain QimenBench-scoped; the cross-source claim concerns the benchmark mechanism.

Related Work

Benchmark saturation motivates designs where the tested capability is explicit and auditable. Recent saturation and item-response analyses argue that static benchmark scores can plateau or become hard to interpret without item-level diagnostics and construction controls (Akhtar et al. 2026; Deveci and Ataman 2025; Zhou et al. 2025). Dynamic evaluation and contamination studies similarly warn that benchmark value depends on source control, leakage analysis, and robust provenance rather than aggregate accuracy alone (Zhu et al. 2024; Wang et al. 2025; Palavalli, Bertsch, and Gormley 2024; Tao et al. 2025). QimenBench follows this construction-first view, but applies it to a compact procedural domain where every reported row has an explicit target-control role.

Classic language-model and reasoning benchmarks provide the lineage for automatic scoring, broad task aggregation, and transparent reporting. The One Billion Word benchmark, MMLU, BIG-bench, HELM, GSM-style verifier work, TruthfulQA, and GPQA each stress a different axis of scale, coverage, exact answer checking, or adversarial item selection (Chelba et al. 2013; Hendrycks et al. 2020; Srivastava et al. 2022; Liang et al. 2022; Cobbe et al. 2021; Lin, Hilton, and Evans 2021; Rein et al. 2023). QimenBench is narrower than these suites, but borrows their insistence that task definitions, scorer behavior, and row counts should be visible to the reader.

Multimodal and domain-specific benchmark papers show how task families and data provenance can be made paper-facing. MMMU, SEED-Bench, MMBench, LVLm-eHub, MME, and DatBench define broad visual or multimodal evaluation surfaces, while LARABench and multilingual safety work illustrate why domain and language boundaries should be named explicitly (Yue et al. 2023; Li et al. 2023; Liu et al. 2023b; Xu et al. 2023; Fu et al. 2023; DatologyAI et al. 2026; Abdelali et al. 2023; Wang et al. 2023b). QimenBench takes the same provenance discipline into a text-only procedural setting and reports F1/F2/F3 as executed components with shared project-created provenance rather than as independent public sources.

Agent, tool-use, and procedural evaluations motivate the need to separate planning, schema following, and hidden derivation failures. AgentBench and Voyager evaluate longer-horizon agent behavior, PPTC-R evaluates task-completion robustness, and SLM-Bench plus task-specific efficiency work compare smaller and larger models under explicit budgets (Liu et al. 2023a; Wang et al. 2023a; Zhang et al. 2024; Pham et al. 2025; Cao et al. 2026). ChaosBench-Logic, causal-reasoning reviews, non-ideal reasoning studies, and prompt-optimization work stress that reasoning claims depend on the exact task regime (Thomas 2026; Yang et al. 2024; Tian et al. 2025; Shah et al. 2026). QimenBench therefore reports a fixed case-study packet rather than a broad model ranking.

Shortcut-learning work is the closest design precedent. VQA shortcut benchmarks and LLM shortcut suites show that systems can exploit correlations or visible cues without learning the intended task, while character-composition and fact-learning studies show how plausible language behavior can hide missing low-level or factual mechanisms (Si et al. 2022; Yuan et al. 2024; Shin and Kaneko 2024; Zucchet et al. 2025). QimenBench makes the shortcut probe part of the benchmark itself: control rows are deliberately copyable, target rows require hidden derivation, and the extraction baseline is expected to pass only the visible side.

Method

QimenBench contains three task families, F1-F3. Each family produces paired items with the same answer schema but different evidence visibility. Control rows expose enough information for copying or extraction to succeed. Target rows require hidden procedural derivation before the same fields can be filled. The exact scorer checks structured fields rather than free-form explanations, so each row is either fully exact or not.

The benchmark construction has four paper-facing stages. First, a task generator creates paired control and target rows. Second, baseline systems produce JSON predictions. Third, an exact scorer compares every required field against the frozen answer key. Fourth, analysis scripts aggregate per-family target-control gaps, Wilson intervals, and failure categories. Figure 2 summarizes this benchmark and evidence pipeline.

The families have different procedural surfaces. F1 focuses on chart-level procedural fields, where a row must recover the dun type, ju number, and xun shou. F2 focuses on duty chief and duty door assignments. F3 uses a larger palace-vector schema, which makes structured-output errors visible in addition to derivation errors. The exact family definitions matter less than the pairing principle: each control row and its target counterpart share the same answer schema, but only the control makes the relevant answer directly visible. That shared schema prevents the evaluation from confusing output-format difficulty with the target-control intervention.

The control rows are not intended to be hard. They are an explicit shortcut probe. If a system cannot solve controls, the benchmark records a basic extraction or schema failure. If it solves controls but fails targets, the benchmark records a derivation failure. If it solves both, the benchmark provides evidence for the corresponding procedural family. This three-way interpretation is the reason the extraction-only baseline is included even though it is not a competitive model.

The scorer is intentionally strict. A row is exact only when all required fields match the frozen answer. Partial field correctness is retained in failure-audit artifacts but does not count as row success. This strictness is appropriate for chart construction because a partially correct chart can be operationally misleading: an incorrect duty door, xun shou, or palace-vector item changes the constructed object. The strict score also makes the oracle sanity run meaningful; 1,440/1,440 exact rows demonstrate that the frozen answer

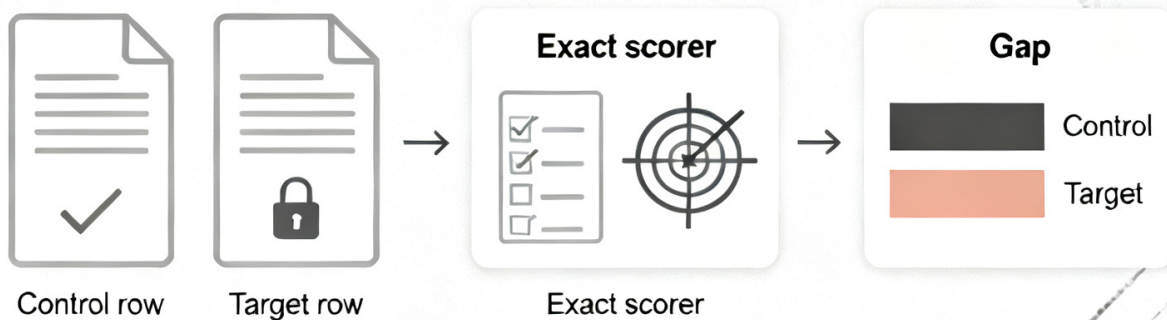


Figure 1: QimenBench pairs visible-answer controls with hidden-derivation targets so shortcut extraction can be separated from procedural chart construction.

packet can round-trip through the same scorer used for baselines.

The method avoids hidden human adjudication. Rows, predictions, exact scores, target-control pairings, and failure categories are represented as structured benchmark records. The manuscript reports aggregate facts from those records and uses row-level audits only to explain scoring and failure categories.

The construction also gives every baseline a falsifiable role. The oracle can fail only if the scorer, answer key, or aggregation layer is inconsistent. The extraction baseline can pass controls but should not be able to derive hidden targets. A model-backed condition can in principle solve either side, but it must be evaluated at enough scale before a model claim is credible. This role separation is useful for a benchmark draft because it makes a failed baseline informative rather than merely disappointing.

The F1-F3 split provides a small but heterogeneous procedural surface. F1 checks compact global fields. F2 checks paired duty assignments. F3 checks a list-like palace vector where the schema itself is harder to satisfy. A shortcut method that copies visible values should pass controls across all three families. The hosted row does exactly that under the fixed prompt path, while the Qwen row also shows basic prompt-following, schema, and control failures. This makes

the current packet a measured negative case study rather than an unexecuted comparison.

The target rows are not adversarial paraphrases of controls. They are matched tasks with the answer visibility changed. That distinction matters. If target rows were simply harder prompts, a target-control gap could reflect prompt length, wording, or schema changes. In QimenBench, the intended intervention is whether the answer can be copied or must be derived. The same exact scorer and output schema then make the gap interpretable as a capability boundary for the tested method.

The benchmark design therefore follows a conservative evaluation principle: a row is not a success because it looks plausible, and a method is not a procedural solver because it can fill visible fields. The evidence packet preserves both aggregate and row-level views so that future reviewers can inspect whether an improvement comes from hidden derivation, better schema following, or a weaker shortcut.

Experimental Setup

The frozen evidence packet evaluates four paper-facing conditions. The oracle sanity condition copies the frozen gold answer and checks scorer/gold consistency. The extraction-only baseline is designed as a shortcut control: it can recover visible controls but has no procedural mechanism for hidden



Figure 2: The QimenBench evaluation pipeline links paired tasks, baselines, exact scoring, gap analysis, and paper evidence.

targets. The hosted condition evaluates ‘gpt-5-mini’ with a JSON-only prompt, temperature 0, a 180-token default output cap, and a 900-token cap for F3 palace-vector rows. The open-weight condition evaluates ‘Qwen/Qwen2.5-7B-Instruct’ with a JSON chat-template prompt, temperature 0, and a 768-token output cap. All four paper-facing conditions have 1,440 scored rows.

Table 1 is the cross-source setup/results matrix. The primary chart-construction source is QimenBench v0.1.0 over F1-F3 paired target/control rows. The task split is 720 controls and 720 targets for each full QimenBench condition, with 240 control and 240 target rows per family. The independent-source packet adds 24 rows each for Gregorian calendar arithmetic, Luhn checksum arithmetic, and Roman numeral transduction, with 12 controls and 12 hidden targets per source. All rows use row-level exact match over required structured fields.

The metric is row-level exact match over the required structured fields. For paired analysis, each target row is matched to its corresponding control row within a family and method. The evidence tables report aggregate results, paired significance, component evidence, failure taxonomy, and model target-failure categories.

Each family has 240 controls and 240 targets per full condition, for 480 rows per family and 1,440 rows across F1-F3. The oracle, extraction baseline, hosted GPT-5 mini, and Qwen conditions all check the same 720 controls and 720 targets under the same exact scorer. Earlier smoke evidence is superseded by the full hosted packet and is not used as paper-facing evidence.

The experiment uses four QimenBench evaluated roles and two independent-source roles. On QimenBench, the oracle tests scorer consistency, extraction tests whether visible-answer copying can be separated from hidden derivation, the hosted role tests one API-backed model under the fixed prompt/scorer path, and the open-weight role tests one instruction-tuned checkpoint under a fixed vLLM backend. On the independent procedural sources, the oracle and visible-answer extractor test whether the paired target-control mechanism generalizes beyond QimenBench provenance before more expensive model-backed cross-source

rows are added.

The paired analysis uses matched control and target rows. For a method and family, the target-control gap is the target exact rate minus the control exact rate. Discordant pairs are then counted as target-only or control-only successes. This pairing is more informative than a single accuracy because it tells the reader whether failures are concentrated on hidden derivation. The extraction baseline is expected to produce control-only discordance if the benchmark construction is doing its job.

The failure analysis uses the saved row-level audits. Extraction failures are summarized by family and condition. Hosted and open-weight target failures are summarized by parse status, schema/type failures, and procedural field mismatches. The model-backed rows therefore support negative case-study claims about the executed conditions while preserving the distinction between procedural failure and output-format failure.

The setup intentionally includes one negative control rather than only model-backed systems. A benchmark that reports only an oracle and a model can tell us whether the model is correct, but not whether the benchmark distinguishes the intended shortcut. The extraction-only baseline fills that role. Its full-packet target failure is not a weakness in the paper; it is the measured behavior the benchmark was designed to expose.

No train/test tuning is performed in this draft. The paper uses the saved task packet and saved prediction rows. That matters because prompt or benchmark iteration after seeing failures could blur the status of the reported rows. The draft treats the existing rows as frozen evidence and describes only external-validation or future-improvement comparisons as future work.

The baseline labels are scientific roles. The oracle is a scorer sanity check, the extraction baseline is a shortcut probe, the hosted run is a fixed-packet model probe, and the open-weight row is an instruction-tuned model probe. The evaluated research object is the benchmark behavior under these roles.

The primary stopping rule for claims is completed scored rows. A planned condition cannot support a performance claim without row-level scores. This rule is why the paper now cites the full hosted and open-weight packets, while leaving external public validation and future solver improvements outside the current result claims.

Results

Table 1 reports the evidence-supported headline numbers for QimenBench and the independent procedural sources. The QimenBench oracle sanity condition is exact on all 1,440 rows, establishing that the frozen answer packet and scorer agree. The extraction-only baseline and hosted GPT-5 mini are each exact on 720 of 1,440 rows. Their aggregate accuracy is therefore 0.500, but the aggregate hides the core diagnostic result: all controls pass and all hidden targets fail. The independent-source oracle is exact on 72/72 rows, while the independent-source shortcut baseline is exact on 36/72 rows, again solving all controls and no hidden targets. Figure 3 visualizes the QimenBench split across families.

Benchmark/ source	Split	Rows	Role	Backend	Budget	Metric	Ctrl	Tgt	Gap	Takeaway
Qimen Bench v0.1.0	F1–F3 full	1,440	Hosted model	gpt-5- mini	JSON; T=0; 180/900	exact row	1.000	0.000	-1.000	chart shortcut
Gregorian calendar	source probe	24	Oracle sanity	rule oracle	det.; no tokens	exact row	1.000	1.000	0.000	source round trip
Gregorian calendar	source probe	24	Shortcut probe	visible extractor	det.; no tokens	exact row	1.000	0.000	-1.000	copying fails hidden
Luhn checksum	source probe	24	Oracle sanity	rule oracle	det.; no tokens	exact row	1.000	1.000	0.000	source round trip
Luhn checksum	source probe	24	Shortcut probe	visible extractor	det.; no tokens	exact row	1.000	0.000	-1.000	copying fails hidden
Roman numerals	source probe	24	Oracle sanity	rule oracle	det.; no tokens	exact row	1.000	1.000	0.000	source round trip
Roman numerals	source probe	24	Shortcut probe	visible extractor	det.; no tokens	exact row	1.000	0.000	-1.000	copying fails hidden

Table 1: Cross-source setup/results matrix: QimenBench v0.1.0 supplies the chart-construction packet, and three independently generated procedural source families test whether the same target-control shortcut signature recurs outside QimenBench provenance. Rows report source, split size, evaluated role/backend, budget, exact-match metric, control rate, target rate, and numerical takeaway.

The per-family pattern is uniform for extraction and hosted GPT-5 mini. F1, F2, and F3 each have 240/240 exact controls and 0/240 exact targets for those two conditions. The oracle condition is also uniform, reaching 240/240 controls and 240/240 targets in each family. These rows separate scorer sanity from shortcut rejection: the oracle rules out a broken answer/scorer packet, and the non-oracle full rows show that visible copying or control solving does not imply hidden derivation. The compact component-level diagnostic appears with the failure discussion in Table 2.

The Qwen2.5-7B-Instruct row has a different failure signature. It solves only 18 controls and no targets across the full packet, with prompt-following, schema, and field failures concentrated in the row-level audit. The paper therefore reports it as one measured open-weight condition, not as a general claim about all open-weight models.

Analysis/Ablation

The paired target-control view is the main ablation. For the extraction-only baseline, each of F1, F2, and F3 has 240 matched pairs, target accuracy 0.000, control accuracy 1.000, and a target-minus-control gap of -1.000. Across the three families, there are 720 control-only discordant pairs

and zero target-only discordant pairs. The exact paired p-value is $1.132e-72$ within each family, showing that the shortcut baseline’s failure is not an aggregate artifact.

The ablation therefore supports the target-control mechanism within QimenBench v0.1.0: the same schema and scoring rule can distinguish copying from derivation. Table 3 gives the matched-pair summary at the transition into the broader failure taxonomy. It also shows that the hosted condition does not beat the nontrivial extraction baseline on hidden targets under the fixed prompt/scorer path.

The case-study packet illustrates why target-control decomposition matters. Extraction and hosted GPT-5 mini have the same 0.500 aggregate, but the paired table shows that this score means 720 control-only discordant pairs, not mixed procedural success. The Qwen result has a much lower aggregate and therefore diagnoses basic schema/control weakness before hidden-derivation ability can be credited.

The failure taxonomy completes the ablation by explaining what the negative rows contain. If all failures were parse failures, QimenBench would mostly test output formatting. If failures were randomly distributed across controls and targets, the target-control intervention would be weak. The

Controls pass while hidden targets fail in each QimenBench family

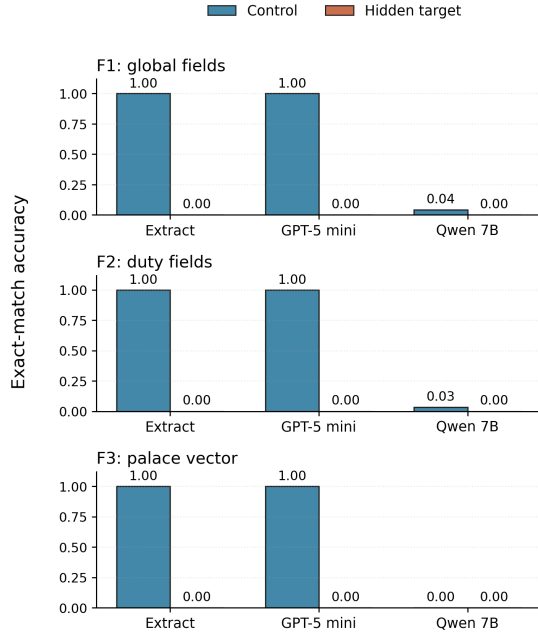


Figure 3: Target-control accuracy exposes the shortcut signature: extraction and hosted GPT-5 mini reach 1.0 on controls and 0.0 on hidden targets across F1-F3; Qwen2.5-7B-Instruct is a measured open-weight negative.

saved audits instead separate visible copying, hidden procedural derivation, and larger-schema errors.

Failure Cases

The failure taxonomy keeps negative rows in the evidence chain. For the extraction-only baseline, target failures are expected: the baseline can copy visible answers but lacks the hidden derivation needed for target fields. For hosted GPT-5 mini, all hidden targets fail while controls pass, so the model-backed row repeats the shortcut signature under the fixed prompt/scorer path. For Qwen2.5-7B-Instruct, the audit shows a stronger collapse: many controls also fail through schema/type and field mismatches.

The family-level failures clarify why exact scoring is useful. F1 and F2 failures can contain plausible-looking procedural fields that do not match the frozen answer. F3 adds schema pressure through palace-vector fields, where malformed or mistyped structures must be separated from wrong but parseable chart values. A free-form inspection could treat these outputs as partially reasonable, but the exact scorer and row-level audit identify which failure mode occurred.

The current packet therefore evaluates benchmark behavior across QimenBench and three independent procedural sources before broad model-ranking claims. A stronger future method should preserve control correctness while improving hidden targets, and it should reduce procedural field mismatches without introducing schema/type errors. Until

model-backed cross-source rows are added, QimenBench should be read as a target-control benchmark construction paper rather than a finished leaderboard.

Scope and Replay Boundary

The current packet should be read as a benchmark-construction study and measured negative result, not as a capability claim about all language models. The executed rows show scorer/gold consistency, reject a visible-copying shortcut, reproduce that shortcut signature for the hosted condition under a fixed QimenBench prompt/scorer path, record one open-weight failure profile, and validate the target-control mechanism on three independent procedural sources.

The source boundary is now split by claim. QimenBench is project-created, which makes the rule packet and scorer auditable but limits broad external validity for Qi Men chart conclusions. The independent procedural packet adds Gregorian calendar, Luhn checksum, and Roman numeral sources, which support the general target-control measurement pattern but are not model-backed leaderboard rows.

The model rows have the same boundary. The hosted row exercises a full prompt, parser, schema, scorer, and audit path at the same 1,440-row scale as the deterministic baselines, but its negative target result is tied to one fixed prompt policy and one benchmark packet. The Qwen row is useful because it records a different failure profile with control and schema weakness, not because it ranks open-weight systems.

The replay contract follows that scope. A reviewer should be able to regenerate the F1-F3 task packet, run or inspect the oracle, extraction, hosted, and open-weight rows, score structured JSON predictions by exact match, and rebuild the aggregate, paired, component, and failure tables. The important identifiers are family labels and target-control pair links; without the pair link, aggregate accuracy would hide the control-only discordance that defines the case study.

The scorer remains outside the model. It receives structured predictions and checks required fields against the frozen answer key, so row correctness is deterministic rather than judged by a language model. Partial field correctness is retained for failure analysis, but the headline metric stays exact match because an incorrect duty door, xun shou, or palace-vector entry changes the constructed chart.

The oracle and extraction conditions are intentionally simple because they answer construction questions. The oracle removes scorer/gold ambiguity. The extraction baseline tests whether target rows leak visible answers. In the frozen packet, the oracle succeeds everywhere and extraction succeeds only on controls, so the benchmark rejects the intended shortcut before stronger solvers are added.

This construction lesson is portable but not yet broad. A future positive solver should cover all F1-F3 pairs under a fixed prompt policy and improve hidden targets while preserving controls. A future external-validation packet should add independent procedural tasks or independently curated Qi Men Dun Jia rows. Those are distinct additions: new sources test benchmark generality, while new models test solver behavior.

Comp.	Diagnostic surface	Extraction	Hosted	Qwen	Scoped provenance note
F1	Dun/Ju plus Xun-Shou boundary	240/0	240/0	10/0	Distinct executed task file and schema; project-created QimenBench provenance.
F2	Duty chief and duty door derivation	240/0	240/0	8/0	Distinct executed task file and schema; project-created QimenBench provenance.
F3	Palace-placement vectors	240/0	240/0	0/0	Distinct executed task file and larger vector schema; project-created QimenBench provenance.

Table 2: Component diagnostic table: each cell reports exact controls/hidden targets out of 240/240 rows. Extraction and hosted GPT-5 mini preserve controls and fail hidden targets in every component, while Qwen2.5-7B-Instruct also shows prompt/schema/control failures.

Method	Pairs	Target	Control	Gap
Oracle sanity	720	1.000	1.000	0.000
Extraction-only	720	0.000	1.000	-1.000
GPT-5 mini hosted	720	0.000	1.000	-1.000
Qwen2.5-7B	720	0.000	0.025	-0.025

Table 3: Paired analysis: extraction and hosted GPT-5 mini each have 720 control-only discordant pairs. The Qwen condition is a measured open-weight negative with 18 exact controls and no exact hidden targets.

The present evidence also defines what would count as progress. A method that raises aggregate accuracy by sacrificing controls would not be a clean procedural improvement. A method that improves compact F1 fields but fails the larger F3 palace-vector schema would be family-specific progress rather than a solved benchmark. A method that succeeds only after prompt iteration would need a fixed prompt policy before comparison.

The figures and tables are scoped to this purpose. The conceptual figures explain the paired construction and evidence flow; they are not additional experiments. The setup/results table states source, split, role, backend, budget, metric, and target-control result for one case-study packet; it is not a leaderboard. The paired table gives the target-control ablation; it is not a robustness study.

The paper therefore keeps completed rows separate from future comparisons. Completed scored rows support result claims; incomplete comparisons support only limitations or future-work statements. This leaves the contribution narrower than a leaderboard announcement, but it makes the negative result auditable: new methods must improve hidden targets, preserve controls, and report field-level failure categories.

The failure categories are part of that audit rather than an extra claim. A target error with a valid schema is different from a malformed output, and both differ from a control failure. QimenBench records those cases separately so that a future improvement can be attributed to hidden derivation, schema control, prompt following, or a mixture of the three. This is why the current negative extraction result is useful: it fixes a known shortcut boundary before stronger systems

are added.

This boundary also explains why the paper does not collapse the evidence into one accuracy number. The extraction-only and hosted rows both produce 0.500 aggregate exact match, but the paired counts show that the score means exact controls and failed hidden targets. The Qwen row has lower aggregate accuracy and a different interpretation because many controls fail too. Treating those rows as the same kind of failure would hide the benchmark behavior that the packet was designed to expose.

The same accounting prevents overclaiming when the packet is expanded. If a future solver closes the hidden-target gap in F1 but not F3, the result should be reported as family-specific progress, not as a solved benchmark. If another model improves targets but introduces schema/type errors, the improvement should be separated from output-format reliability. If independent sources are added, the source label should remain visible in the results matrix so readers can distinguish benchmark generality from solver behavior.

Finally, the practical scope is deliberately limited. The paper does not evaluate deployment decisions, user-facing advice, or cultural authority in Qi Men Dun Jia. It uses the domain as a compact procedural setting with fixed structured answers. The main risk is that a reader might overgeneralize a scoped construction result into a broad model-capability statement. The manuscript addresses that risk by naming completed rows, retaining the shared-provenance caveat, and separating future source and model expansions from current claims.

Conclusion

QimenBench is a benchmark-construction study for procedural chart tasks with cross-source target-control validation. Existing evidence supports one main observation: paired target/control construction exposes shortcut behavior that an aggregate exact-match score can hide. The frozen QimenBench oracle, extraction, hosted GPT-5 mini, and Qwen2.5-7B-Instruct runs are complete for the chart packet, and the independent procedural packet shows the same visible-copying signature across Gregorian calendar, Luhn checksum, and Roman numeral sources.

The useful outcome is therefore methodological rather than leaderboard-oriented. QimenBench shows how a

benchmark packet can make a negative result interpretable: controls establish visible-answer and schema behavior, targets test hidden derivation under the same scorer, and paired accounting distinguishes shortcut success from procedural success. Future work should keep that accounting fixed while adding model-backed rows on the independent sources and stronger solvers.

This separation is the main submission boundary. The current manuscript can support claims about the construction, the frozen scorer, the target-control signature, cross-source shortcut validation, and measured negative QimenBench rows. It cannot yet support a claim that a class of models broadly fails every procedural source family, because the independent-source expansion currently uses oracle and shortcut baselines rather than API/open-weight model rows. That boundary leaves the next comparison falsifiable instead of implied.

References

- Abdelali, A.; Mubarak, H.; Chowdhury, S. A.; Hasanain, M.; Boughorbel, S.; Abdaljalil, S.; Ahmad, B.; Nazar, N.; Boudabous, Y.; Al-Twairish, N.; Alqwaifly, S.; Almazrouei, E.; Tuffaha, H.; Muhammad, S. H.; Gargouri, B.; Abbas, M.; Khelifi, L.; Zrigui, M.; Hamdi, T.; Habash, N.; and Alam, F. 2023. LARA-Bench: Benchmarking Arabic AI with Large Language Models. *arXiv preprint*.
- Akhtar, M.; Reuel, A.; Soni, P.; Ahuja, S.; Ammanamanchi, P. S.; Rawal, R.; Zouhar, V.; Yadav, S.; Whitehouse, C.; Ki, D.; Mickel, J.; Choshen, L.; Suppa, M.; Batzner, J.; Chim, J.; Sania, J.; Long, Y.; Rahmani, H. A.; Knight, C.; Nan, Y.; Raj, J.; Fan, Y.; Singh, S.; Sahoo, S.; Habba, E.; Gohar, U.; Pawar, S.; Scholz, R.; Subramonian, A.; Ni, J.; Kochenderfer, M.; Koyejo, S.; Sachan, M.; Biderman, S.; Talat, Z.; Ghosh, A.; and Solaiman, I. 2026. When AI Benchmarks Plateau: A Systematic Study of Benchmark Saturation. *arXiv preprint*.
- Cao, J.; Ma, Y.; Li, X.; Ren, Q.; and Ma, M. 2026. Task-Specific Efficiency Analysis: When Small Language Models Outperform Large Language Models. *arXiv preprint*.
- Chelba, C.; Mikolov, T.; Schuster, M.; Ge, Q.; Brants, T.; and Koehn, P. 2013. One Billion Word Benchmark for Measuring Progress in Statistical Language Modeling. *arXiv preprint*.
- Cobbe, K.; Kosaraju, V.; Bavarian, M.; Chen, M.; Jun, H.; Kaiser, L.; Plappert, M.; Tworek, J.; Hilton, J.; Nakano, R.; Hesse, C.; and Schulman, J. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint*.
- DatologyAI; ; Joshi, S.; Yin, H.; Adiga, R.; Monti, R.; Caranza, A.; Fang, A.; Deng, A.; Abbas, A.; Larsen, B.; Blakeney, C.; Teh, D.; Schwab, D.; Pan, F.; Mongstad, H.; Urbanek, J.; Lee, J.; Telanoff, J.; Wills, J.; Mentzer, K.; Merrick, L.; Doshi, P.; Burstein, P.; Maini, P.; Loftin, S.; Das, S.; Jiang, T.; Dorna, V.; Wang, Z.; Giza, B.; Morcos, A.; and Leavitt, M. 2026. DatBench: Discriminative, Faithful, and Efficient VLM Evaluations. *arXiv preprint*.
- Deveci, I. E.; and Ataman, D. 2025. The Ouroboros of Benchmarking: Reasoning Evaluation in an Era of Saturation. *arXiv preprint*.
- Fu, C.; Chen, P.; Shen, Y.; Qin, Y.; Zhang, M.; Lin, X.; Yang, J.; Zheng, X.; Li, K.; Sun, X.; Wu, Y.; and Ji, R. 2023. MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models. *arXiv preprint*.
- Hendrycks, D.; Burns, C.; Basart, S.; Zou, A.; Mazeika, M.; Song, D.; and Steinhardt, J. 2020. Measuring Massive Multitask Language Understanding. *ICLR 2021 / arXiv*.
- Li, B.; Wang, R.; Wang, G.; Ge, Y.; Ge, Y.; and Shan, Y. 2023. SEED-Bench: Benchmarking Multimodal LLMs with Generative Comprehension. *arXiv preprint*.
- Liang, P.; Bommasani, R.; Lee, T.; Tsipras, D.; Soylu, D.; Yasunaga, M.; Zhang, Y.; Narayanan, D.; Wu, Y.; Kumar, A.; Newman, B.; Yuan, B.; Yan, B.; Zhang, C.; Cosgrove, C.; Manning, C. D.; Re, C.; Acosta-Navas, D.; Hudson, D. A.; Zelikman, E.; Durmus, E.; Ladhak, F.; Rong, F.; Ren, H.; Yao, H.; Wang, J.; Santhanam, K.; Orr, L.; Zheng, L.; Yuksekgonul, M.; Suzgun, M.; Kim, N.; Guha, N.; Chatterji, N.; Khattab, O.; Henderson, P.; Huang, Q.; Chi, R.; Xie, S. M.; Santurkar, S.; Ganguli, S.; Hashimoto, T.; Icard, T.; Zhang, T.; Chaudhary, V.; Wang, W.; Li, X.; Mai, Y.; Zhang, Y.; and Koreeda, Y. 2022. Holistic Evaluation of Language Models. *arXiv preprint*.
- Lin, S.; Hilton, J.; and Evans, O. 2021. TruthfulQA: Measuring How Models Mimic Human Falsehoods. *ACL 2022 / arXiv*.
- Liu, X.; Yu, H.; Zhang, H.; Xu, Y.; Lei, X.; Lai, H.; Gu, Y.; Ding, H.; Men, K.; Yang, K.; Zhang, S.; Deng, X.; Zeng, A.; Du, Z.; Zhang, C.; Shen, S.; Zhang, T.; Su, Y.; Sun, H.; Huang, M.; Dong, Y.; and Tang, J. 2023a. AgentBench: Evaluating LLMs as Agents. *arXiv preprint*.
- Liu, Y.; Duan, H.; Zhang, Y.; Li, B.; Zhang, S.; Zhao, W.; Yuan, Y.; Wang, J.; He, C.; Liu, Z.; Chen, K.; and Lin, D. 2023b. MMBench: Is Your Multi-modal Model an All-around Player? *arXiv preprint*.
- Palavalli, M.; Bertsch, A.; and Gormley, M. R. 2024. A Taxonomy for Data Contamination in Large Language Models. *arXiv preprint*.
- Pham, N. T.; Kieu, T.; Nguyen, D.-M.; Xuan, S. H.; Nguyen, A. G.-T.; Pham, H.; Tran, M. Q.; Pham, P. T.; Phung, D.; Tran, B.; and Le, T. 2025. SLM-Bench: A Comprehensive Benchmark of Small Language Models on Environmental Impacts—Extended Version. *arXiv preprint*.
- Rein, D.; Hou, B. L.; Stickland, A. C.; Petty, J.; Pang, R. Y.; Dirani, J.; Michael, J.; and Bowman, S. R. 2023. GPQA: A Graduate-Level Google-Proof Q&A Benchmark. *arXiv preprint*.
- Shah, D.; Badhe, S.; Kathrotia, N.; Tiwari, P.; and Verma, R. 2026. CROP: Token-Efficient Reasoning in Large Language Models via Regularized Prompt Optimization. *arXiv preprint*.
- Shin, A.; and Kaneko, K. 2024. Large Language Models Lack Understanding of Character Composition of Words. *arXiv preprint*.
- Si, Q.; Meng, F.; Zheng, M.; Lin, Z.; Liu, Y.; Fu, P.; Cao, Y.; Wang, W.; and Zhou, J. 2022. Language Prior Is Not the Only Shortcut: A Benchmark for Shortcut Learning in VQA.

- Srivastava, A.; Rastogi, A.; Rao, A.; Shueb, A. A. M.; Abid, A.; Fisch, A.; Brown, A. R.; Santoro, A.; Gupta, A.; Garriga-Alonso, A.; Kluska, A.; Lewkowycz, A.; Agarwal, A.; Power, A.; Ray, A.; Warstadt, A.; Kocurek, A. W.; Safaya, A.; Tazarv, A.; Xiang, A.; Parrish, A.; Nie, A.; Hussain, A.; Askell, A.; Dsouza, A.; Slone, A.; Rahane, A.; Iyer, A. S.; Andreassen, A.; Madotto, A.; Santilli, A.; Stuhlmuller, A.; Dai, A.; La, A.; Lampinen, A.; Zou, A.; Jiang, A.; Chen, A.; Vuong, A.; Gupta, A.; Gottardi, A.; Norelli, A.; Venkatesh, A.; Gholamidavoodi, A.; Tabassum, A.; Menezes, A.; Kirubarajan, A.; Mullokandov, A.; Sabharwal, A.; Herrick, A.; Efrat, A.; Erdem, A.; Karakas, A.; Roberts, B. R.; Loe, B. S.; Zoph, B.; Bojanowski, B.; Ozyurt, B.; Hedayatnia, B.; Neyshabur, B.; Inden, B.; Stein, B.; Ekmekci, B.; Lin, B. Y.; Howald, B.; Orinon, B.; Diao, C.; Dour, C.; Stinson, C.; Argueta, C.; Ramirez, C. F.; Singh, C.; Rathkopf, C.; Meng, C.; Baral, C.; Wu, C.; Callison-Burch, C.; Waites, C.; Voigt, C.; Manning, C. D.; Potts, C.; Ramirez, C.; Rivera, C. E.; Siro, C.; Raffel, C.; Ashcraft, C.; Garbacea, C.; Sileo, D.; Garrette, D.; Hendrycks, D.; Kilman, D.; Roth, D.; Freeman, D.; Khashabi, D.; Levy, D.; Gonzalez, D. M.; Perszyk, D.; Hernandez, D.; Chen, D.; Ippolito, D.; Gilboa, D.; Dohan, D.; Drakard, D.; Jurgens, D.; Datta, D.; Ganguli, D.; Emelin, D.; Kleyko, D.; Yuret, D.; Chen, D.; Tam, D.; Hupkes, D.; Misra, D.; Buzan, D.; Molloy, D. C.; Yang, D.; Lee, D.-H.; Schrader, D.; Shutova, E.; Cubuk, E. D.; Segal, E.; Hagerman, E.; Barnes, E.; Donoway, E.; Pavlick, E.; Rodola, E.; Lam, E.; Chu, E.; Tang, E.; Erdem, E.; Chang, E.; Chi, E. A.; Dyer, E.; Jerzak, E.; Kim, E.; Manyasi, E. E.; Zheltonozhskii, E.; Xia, F.; Siar, F.; Martinez-Plumed, F.; Happe, F.; Chollet, F.; Rong, F.; Mishra, G.; Winata, G. I.; de Melo, G.; Kruszewski, G.; Parascandolo, G.; Mariani, G.; Wang, G.; Jaimovitch-Lopez, G.; Betz, G.; Gur-Ari, G.; Galijasevic, H.; Kim, H.; Rashkin, H.; Hajishirzi, H.; Mehta, H.; Bogar, H.; Shevlin, H.; Schutze, H.; Yakura, H.; Zhang, H.; Wong, H. M.; Ng, I.; Noble, I.; Jumelet, J.; Geissinger, J.; Kernion, J.; Hilton, J.; Lee, J.; Fisac, J. F.; Simon, J. B.; Koppel, J.; Zheng, J.; Zou, J.; Koccon, J.; Thompson, J.; Wingfield, J.; Kaplan, J.; Radom, J.; Sohl-Dickstein, J.; Phang, J.; Wei, J.; Yosinski, J.; Novikova, J.; Bosscher, J.; Marsh, J.; Kim, J.; Taal, J.; Engel, J.; Alabi, J.; Xu, J.; Song, J.; Tang, J.; Waweru, J.; Burden, J.; Miller, J.; Balis, J. U.; Batchelder, J.; Berant, J.; Frohberg, J.; Rozen, J.; Hernandez-Orallo, J.; Boudeman, J.; Guerr, J.; Jones, J.; Tenenbaum, J. B.; Rule, J. S.; Chua, J.; Kanclerz, K.; Livescu, K.; Krauth, K.; Gopalakrishnan, K.; Ignatyeva, K.; Markert, K.; Dhole, K. D.; Gimpel, K.; Omondi, K.; Mathewson, K.; Chiafullo, K.; Shkaruta, K.; Shridhar, K.; McDonell, K.; Richardson, K.; Reynolds, L.; Gao, L.; Zhang, L.; Dugan, L.; Qin, L.; Contreras-Ochando, L.; Morency, L.-P.; Moschella, L.; Lam, L.; Noble, L.; Schmidt, L.; He, L.; Colon, L. O.; Metz, L.; Senel, L. K.; Bosma, M.; Sap, M.; ter Hoeve, M.; Farooqi, M.; Faruqui, M.; Mazeika, M.; Baturan, M.; Marelli, M.; Maru, M.; Quintana, M. J. R.; Tolkiehn, M.; Giulianelli, M.; Lewis, M.; Potthast, M.; Leavitt, M. L.; Hagen, M.; Schubert, M.; Baitemirova, M. O.; Arnaud, M.; McElrath, M.; Yee, M. A.; Cohen, M.; Gu, M.; Ivanitskiy, M.; Starritt, M.; Strube, M.; Swedrowski, M.; Bevilacqua, M.; Yasunaga, M.; Kale, M.; Cain, M.; Xu, M.; Suzgun, M.; Walker, M.; Tiwari, M.; Bansal, M.; Aminnaseri, M.; Geva, M.; Gheini, M.; T, M. V.; Peng, N.; Chi, N. A.; Lee, N.; Krakover, N. G.-A.; Cameron, N.; Roberts, N.; Doiron, N.; Martinez, N.; Nangia, N.; Deckers, N.; Muennighoff, N.; Keskar, N. S.; Iyer, N. S.; Constant, N.; Fiedel, N.; Wen, N.; Zhang, O.; Agha, O.; Elbaghdadi, O.; Levy, O.; Evans, O.; Casares, P. A. M.; Doshi, P.; Fung, P.; Liang, P. P.; Vicol, P.; Alipoormolabashi, P.; Liao, P.; Liang, P.; Chang, P.; Eckersley, P.; Htut, P. M.; Hwang, P.; Milkowski, P.; Patil, P.; Pezeshkpour, P.; Oli, P.; Mei, Q.; Lyu, Q.; Chen, Q.; Banjade, R.; Rudolph, R. E.; Gabriel, R.; Habacker, R.; Risco, R.; Milliere, R.; Garg, R.; Barnes, R.; Saurous, R. A.; Arakawa, R.; Raymaekers, R.; Frank, R.; Sikand, R.; Novak, R.; Sitelew, R.; LeBras, R.; Liu, R.; Jacobs, R.; Zhang, R.; Salakhutdinov, R.; Chi, R.; Lee, R.; Stovall, R.; Teehan, R.; Yang, R.; Singh, S.; Mohammad, S. M.; Anand, S.; Dillavou, S.; Shleifer, S.; Wiseman, S.; Gruetter, S.; Bowman, S. R.; Schoenholz, S. S.; Han, S.; Kwatra, S.; Rous, S. A.; Ghazarian, S.; Ghosh, S.; Casey, S.; Bischoff, S.; Gehrmann, S.; Schuster, S.; Sadeghi, S.; Hamdan, S.; Zhou, S.; Srivastava, S.; Shi, S.; Singh, S.; Asaadi, S.; Gu, S. S.; Pachchigar, S.; Toshniwal, S.; Upadhyay, S.; Shyamolima; Debnath; Shakeri, S.; Thormeyer, S.; Melzi, S.; Reddy, S.; Makini, S. P.; Lee, S.-H.; Torene, S.; Hatwar, S.; Dehaene, S.; Divic, S.; Ermon, S.; Biderman, S.; Lin, S.; Prasad, S.; Piantadosi, S. T.; Shieber, S. M.; Mishnerghi, S.; Kiritchenko, S.; Mishra, S.; Linzen, T.; Schuster, T.; Li, T.; Yu, T.; Ali, T.; Hashimoto, T.; Wu, T.-L.; Desbordes, T.; Rothschild, T.; Phan, T.; Wang, T.; Nkinyili, T.; Schick, T.; Kornev, T.; Tunduny, T.; Gerstenberg, T.; Chang, T.; Neeraj, T.; Khot, T.; Shultz, T.; Shaham, U.; Misra, V.; Demberg, V.; Nyamai, V.; Raunak, V.; Ramasesh, V.; Prabhu, V. U.; Padmakumar, V.; Srikumar, V.; Fedus, W.; Saunders, W.; Zhang, W.; Vossen, W.; Ren, X.; Tong, X.; Zhao, X.; Wu, X.; Shen, X.; Yaghoobzadeh, Y.; Lakretz, Y.; Song, Y.; Bahri, Y.; Choi, Y.; Yang, Y.; Hao, Y.; Chen, Y.; Belinkov, Y.; Hou, Y.; Hou, Y.; Bai, Y.; Seid, Z.; Zhao, Z.; Wang, Z.; Wang, Z. J.; Wang, Z.; and Wu, Z. 2022. Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models. *TMLR 2023 / arXiv*.
- Tao, Y.; Wang, T.; Dong, Y.; Liu, H.; Huo, Y.; Peng, B.; Wen, Q.; Wang, Z.; Xu, W.; and Zhang, Y. 2025. Detecting Data Contamination from Reinforcement Learning Post-training for Large Language Models. *arXiv preprint*.
- Thomas, N. 2026. ChaosBench-Logic: A Benchmark for Logical and Symbolic Reasoning on Chaotic Dynamical Systems. *arXiv preprint*.
- Tian, C.; Blaschko, M. B.; Xing, M.; and Li, X. 2025. Large Language Models Reasoning Abilities Under Non-Ideal Conditions After RL-Fine-Tuning. *arXiv preprint*.
- Wang, G.; Xie, Y.; Jiang, Y.; Mandelkar, A.; Xiao, C.; Zhu, Y.; Fan, L.; and Anandkumar, A. 2023a. Voyager: An Open-Ended Embodied Agent with Large Language Models. *arXiv preprint*.
- Wang, H.; Li, H.; Ko, B.; and Zhang, H. 2025. On The

Fragility of Benchmark Contamination Detection in Reasoning Models. *arXiv preprint*.

Wang, W.; Tu, Z.; Chen, C.; Yuan, Y.; tse Huang, J.; Jiao, W.; Wang, W.; tse Huang, J.; Huang, F.; Wang, X.; Shi, S.; and Tu, Z. 2023b. All Languages Matter: On the Multilingual Safety of Large Language Models. *arXiv preprint*.

Xu, P.; Shao, W.; Zhang, K.; Gao, P.; Liu, S.; Cao, M.; Lei, W.; Meng, F.; Huang, S.; Qiao, Y.; and Luo, P. 2023. LVLMeHub: A Comprehensive Evaluation Benchmark for Large Vision-Language Models. *arXiv preprint*.

Yang, L.; Shirvaikar, V.; Clivio, O.; and Falck, F. 2024. A Critical Review of Causal Reasoning Benchmarks for Large Language Models. *arXiv preprint*.

Yuan, Y.; Zhao, L.; Zhang, K.; Zheng, G.; and Liu, Q. 2024. Do LLMs Overcome Shortcut Learning? An Evaluation of Shortcut Challenges in Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.

Yue, X.; Ni, Y.; Zhang, K.; Zheng, T.; Liu, R.; Zhang, G.; Stevens, S.; Jiang, D.; Ren, W.; Sun, Y.; Wei, C.; Yu, B.; Yuan, R.; Sun, R.; Yin, M.; Zheng, B.; Yang, Z.; Liu, Y.; Huang, W.; Sun, H.; Su, Y.; and Chen, W. 2023. MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI. *arXiv preprint*.

Zhang, Z.; Guo, Y.; Liang, Y.; Zhao, D.; and Wan, X. 2024. PPTC-R Benchmark: Towards Evaluating the Robustness of Large Language Models for PowerPoint Task Completion. *arXiv preprint*.

Zhou, H.; Huang, H.; Zhao, Z.; Han, L.; Wang, H.; Chen, K.; Yang, M.; Bao, W.; Dong, J.; Xu, B.; Zhu, C.; Cao, H.; and Zhao, T. 2025. Lost in Benchmarks? Rethinking Large Language Model Benchmarking with Item Response Theory. *arXiv preprint*.

Zhu, K.; Wang, J.; Zhao, Q.; Xu, R.; and Xie, X. 2024. Dynamic Evaluation of Large Language Models by Meta Probing Agents. *ICML 2024 / arXiv*.

Zucchet, N.; Bornschein, J.; Chan, S.; Lampinen, A.; Pascanu, R.; and Garriga-Alonso, A. 2025. How do Language Models Learn Facts? Dynamics, Curricula and Hallucinations. *arXiv preprint*.

Reproducibility Checklist

This draft uses an existing frozen evidence packet. The public-facing replay contract is: regenerate QimenBench F1-F3 tasks, run or inspect the oracle, extraction-only, hosted GPT-5 mini, and Qwen2.5-7B-Instruct prediction rows, score structured JSON fields with exact match, and rebuild the aggregate result, paired-significance, component-evidence, and failure-summary tables. The replay package records seed policy, task versions, row counts, and exact-match scoring rules. No training, finetuning, or new model baseline is required for the reported draft evidence.

Appendix: Additional Evidence Notes

The appendix records scope boundaries and detailed interpretation notes that are not main-body claims. The hosted

GPT-5 mini row is full-scale negative evidence under the fixed prompt/scorer path, and the Qwen2.5-7B-Instruct row is a measured open-weight negative. They are not converted into a broad leaderboard or universal model-capability claim. Additional replay materials should preserve the same task families, exact scorer, paired analysis, and failure-summary format.

Future hosted and open-weight runs should report the same control rate, target rate, target-control gap, paired discordance counts, and failure categories. Adding more models without preserving controls would weaken the diagnostic value. Adding more prompts without row-level failure audits would make it harder to tell whether improvements came from derivation or formatting.

QimenBench is project-created, which limits broad external-validity claims but makes the current construction claim auditable. The F1/F2/F3 component bundles are executed and separately packaged, but they share the same rule manifest/provenance. The task generator, answer schemas, scorer, and baseline/model outputs are available as artifacts. The next evidence stage should add independent curation and stronger positive solvers while preserving paired target-control accounting.